

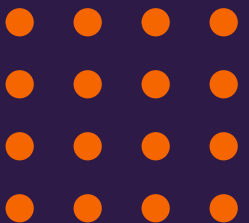


Ethics of Data Science

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A couple clarifications

- Ethics is:
 - The principles of conduct governing an individual or a group
 - The discipline dealing with what is good and bad with moral duty and obligation
- Ethics is not:
 - The law
 - A checklist that can be finished
 - Optional to engage with



Ethical principles for data science



Transparency



Fairness



Accountability



Privacy and security



Many more!



ACM code of ethics (The Code)

<https://www.acm.org/code-of-ethics>

- General ethical principles
- Professional Responsibilities
- Professional Leadership Principles
- Compliance with the Code



Data Science Lifecycle

- Data collection
 - Consenting and informed participation
 - Ownership and origin of the data
- Data processing
 - Handling of sensitive information
 - Biases in data cleaning and aggregation
- Model development and training
 - Bias in algorithms and data
- Deployment and usage
 - Explainability and transparency
 - Risks of unintended uses



Case Study #1 – Facial Recognition Technology

- Biometric system that identifies or verifies individuals by analyzing facial features
- Used in lots of fields, not just unlocking phones/laptops
 - Law enforcement
 - Security
 - Marketing
 - Healthcare



Facial Recognition Technology Issues

- Privacy
 - Who gets their face scanned, where, and for what reason?
- Bias & Discrimination
 - Can the model identify every face with the same accuracy?
(This has so far been an issue with non-white faces)
- Accountability
 - Who is to blame when somebody is wrongly incarcerated?



Case Study #2 – Targeted Advertising

- A marketing strategy using data (e.g. browsing history, location, social media activity, demographics) to display personalized ads
- Commonly used by platforms like Google, Facebook, Amazon, etc.



Targeted Advertising Issues

- Informed consent
 - Do the users know what data is being collected and by who?
- Algorithmic audits for bias
 - Who checks the algorithms to make sure the algorithms are not suggesting based on stereotypes?
- Opt-out Options
 - Is it simple for a user to decide not to take part?
 - Should companies be required to offer opting-out?



Two Organizations to Know



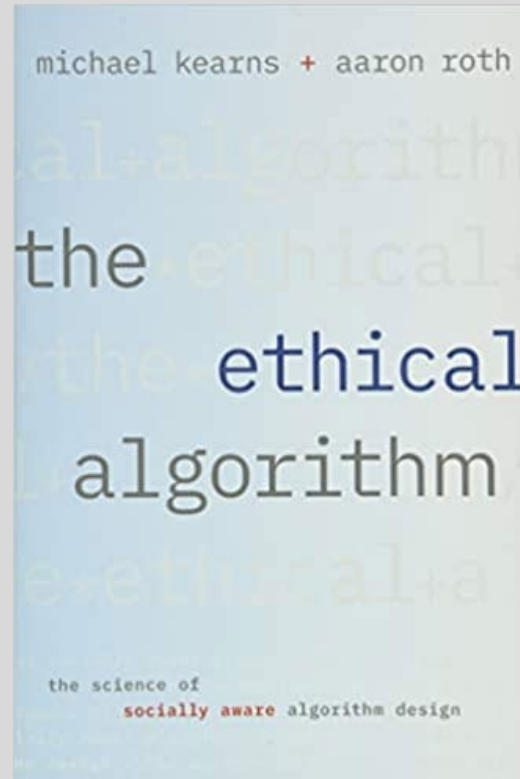
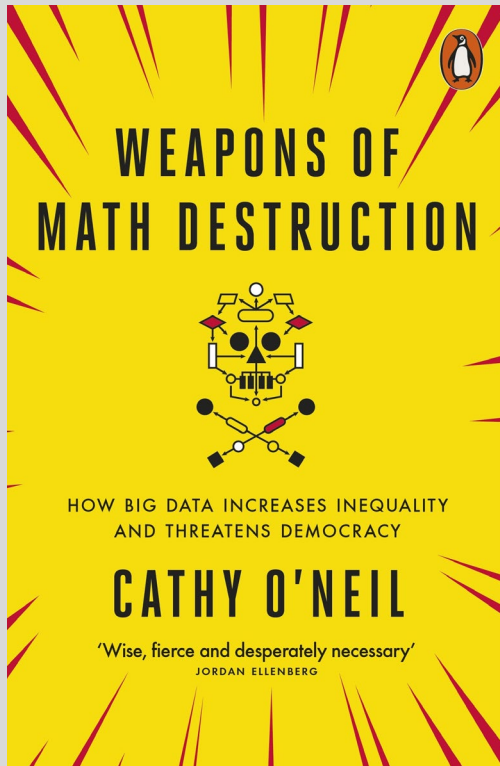
The Algorithmic Justice League – www.ajl.org



The Electronic Frontier Foundation – www.eff.org

Further reading

- Weapons of Math Destruction by Cathy O'Neil
- The Ethical Algorithm by Aaron Roth and Michael Kearns



Questions?